Sidharth Dinesh

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Education

Arizona State University, Tempe, AZ — Aug 2022 – May 2024
Master of Computer Science — GPA: 3.90/4.00

Visvesvaraya National Institute of Technology, Nagpur, India — Jul 2018 – May 2022
BTech, Electronics and Communication Engineering — CGPA: 8.69/10.00
Department rank: 12 of 141 students

Research Experience

Visvesvaraya National Institute of Technology — Nagpur, India
Undergraduate Thesis Research — Jul 2021 – May 2022

Advisor: Prof. Ashwin G. Kothari
A Cost-effective Edge Computing Framework for Video Processing

- Developed workload redistribution logic to balance video processing across edge servers.
- Used video-buffer utilization to identify overloaded and underused edge servers.
- Implemented server pairing to transfer video frames according to available processing capacity.
- Compared processing latency and server workloads against an existing offloading method.

Publications

Prasanna Deshpande, **Sidharth Dinesh**, Apurva Umredkar, and Ashwin G. Kothari. "Improved search and rescue optimization based load balancing scheme of CCTV footage on edge computing environment." *Concurrency and Computation: Practice and Experience*, 35(5), e7556, 2023. [DOI] [PDF] [Repository]

Selected Projects

Spatiotemporal Hotspot Detection in NYC Taxi Data [Repository]

- Implemented spatial joins in Scala and Spark SQL to measure taxi pickup density across regions.
- Mapped pickup coordinates and timestamps to a geographic grid with daily time intervals.
- Computed Getis–Ord G_i^* scores to identify unusually high pickup activity across nearby cells.
- Used distributed aggregations and cached cell counts to rank the top 50 location–day hotspots.

Interactive Map of Dinosaur Records [Repository]

- Plotted 309 dinosaur records on an interactive world map linked to a geological timeline using D3.js.
- Implemented a draggable time selector to explore geographic distributions across geological periods.
- Used color to distinguish dinosaur types and animated map updates across locations and time.

Student Recognition from Classroom Videos [Repository]

- Implemented Python handlers for video recognition on AWS Lambda and OpenFaaS.
- Retrieved student details from DynamoDB after face recognition and saved CSV results to Amazon S3.
- Tested the video-processing pipeline with 100 simultaneous uploads and verified the generated results.

Facebook Comment-Volume Prediction [Repository]

- Evaluated Lasso, gradient boosting, random forest, and neural networks for comment prediction.
- Scaled input features and applied PCA to reduce the prediction inputs to four components.
- Compared prediction error and evaluation time across five variants of Facebook activity data.

Professional Experience

W. P. Carey School of Business, Arizona State University Tempe, AZ
Systems Analyst Apr 2024 – Present

- Developed AdvisingSOS platform for undergraduate advising using React, Node.js, and SQL Server.
- Integrated SQL Server records with university data warehouse for advising and institutional reports.
- Configured Qualtrics assessments with automated scoring and reports for students and instructors.
- Helped building and integrating an AI chatbot into a simulation-based learning platform through APIs.
- Automated internship applications, status tracking, and faculty emails using Airtable.
- Built a Tableau dashboard that gave departments a shared view of the school's corporate partnerships.

Admission Services, Arizona State University Tempe, AZ
Salesforce Data Analyst, Student Worker Jul 2023 – Apr 2024

- Supported Salesforce integrations and STOVA event management workflows for recruitment operations.
- Used SQL to review institutional records and investigate data inconsistencies across systems.
- Cleaned and validated Salesforce admissions records, resolving duplicates and inconsistent data.

College of Health Solutions, Arizona State University Downtown Phoenix, AZ
Office Aide, Student Success Hub Mar 2023 – Aug 2023

- Supported student advising and recruitment by managing Salesforce cases, and appointments in Excel.

IoTIoT.in Pune, India
Software Engineering Intern Sep 2020 – Dec 2020

- Programmed a six-channel relay in C++ for remote control on a Linux-based Rock Pi S.
- Parsed JSON commands received over MQTT and used libcurl for remote relay control.
- Led three collaborators in debugging and testing live CCTV processing in Docker containers.
- Used Shunya AI APIs to detect people and objects, and stored detection timestamps in InfluxDB.

Teaching and Mentoring

Academic Support Network, Arizona State University Tempe, AZ
LEAD MAT 117 Tutor, Student Worker Aug 2023 – Nov 2023

- Tutored MAT 117 students in mathematical concepts and guided problem-solving through ALEKS.

Visvesvaraya National Institute of Technology Nagpur, India
Student Mentor Jul 2020 – May 2022

- Mentored 15 first-year students on academics, campus life, and online learning during COVID-19.

Presentations and Recognition

- Presented on ASU data and Airtable at the WPC Professional Development Conference.
- Led a staff AI session on ASU tools, effective prompting, and responsible data use.
- Received three Sun Awards for the conference presentation and staff AI training.

Technical Skills

Languages: Python, SQL, JavaScript, Scala, C++

Data Systems: PostgreSQL, SQL Server, Spark, Amazon Redshift, DynamoDB, Kafka, InfluxDB

Cloud and Development: AWS (EC2, S3, Lambda, ECR), Docker, Git, Linux, Node.js, React

Data Visualization: D3.js, Tableau, Power BI